Biodivers Conserv (2017) 26:1–21
DOI 10.1007/s10531-016-1223-6

REVIEW PAPER

The Capercaillie (*Tetrao urogallus*): an iconic focal species for knowledge-based integrative management and conservation of Baltic forests

Asko Lõhmus[1] · Meelis Leivits[2] · Elmārs Pēterhofs[3] · Rytis Zizas[4] · Helmuts Hofmanis[5] · Ivar Ojaste[6,7] · Petras Kurlavičius[8]

Received: 18 April 2016 / Revised: 1 September 2016 / Accepted: 24 September 2016 / Published online: 17 October 2016

Abstract Biodiversity loss was a central argument for redefining sustainable forest management in the 1990s, but threatened species remain poorly addressed in forestry governance. Management history of the Capercaillie (*Tetrao urogallus*) population in the Baltic States reveals a high potential of socially valued threatened species for developing the missing forestry–conservation interfaces. We review the history of the Baltic Capercaillie population since the 19th century, showing how its status transformed, both ecologically and socially, from a famous hunting target to the most widely protected forest

Communicated by Matts Lindbladh.

This article belongs to the Topical Collection: Forest and plantation biodiversity.

Electronic supplementary material The online version of this article (doi:10.1007/s10531-016-1223-6) contains supplementary material, which is available to authorized users.

✉ Asko Lõhmus
asko.lohmus@ut.ee

1 Department of Zoology, Institute of Ecology and Earth Sciences, University of Tartu, Vanemuise 46, 51014 Tartu, Estonia

2 Department of Geography, Institute of Ecology and Earth Sciences, University of Tartu, Vanemuise 46, 51014 Tartu, Estonia

3 Latvijas valsts meži, Vaiņodes 1, Riga 1004, Latvia

4 Institute of Forestry, Lithuanian Research Centre for Agriculture and Forestry, Liepu st. 1, Kaunas District, 53101 Girionys, Lithuania

5 Nature Conservation Agency of Latvia, Baznīcas iela 7, Sigulda 2150, Latvia

6 Department of Landscape Management and Nature Conservation, Institute of Agricultural and Environmental Sciences, Estonian University of Life Sciences, Kreutzwaldi 5, 51014 Tartu, Estonia

7 Estonian Ornithological Society, Veski 4, 51005 Tartu, Estonia

8 Lithuanian University of Educational Sciences, Studentu st. 39, 08106 Vilnius, Lithuania

species in the region. Compilation of recent national survey data confirms that at least 3450 lekking males currently survive in 961 leks; they are distributed between six large and about twenty small populations. During the 20th century, lek sizes decreased and local extirpations spread from South-Baltic mosaic lands to northern forests. As a social response, innovative management initiatives have repeatedly enabled periods of population stability and local recoveries. The most recent developments in Capercaillie conservation combine elements from the historically separated nature conservation, forestry and game-management approaches. The consistency of such social responses despite political upheavals suggests that iconic species can culturally stabilize long-term sustainable development.

Keywords Conservation history · Flagship species · Game management · Knowledge system · Population decline · Sustainable forest management

Introduction

Biodiversity has been increasingly accepted as integral for sustainable economy since the 1992 Earth Summit in Rio de Janeiro, but the global progress from the acceptance to implementation has been slow (Gasparatos and Willis 2015). A pioneering approach was taken in the European forest policy, which succeeded to elaborate a common set of criteria and indicators through a pan-continental policy platform within a decade (Rametsteiner and Mayer 2004). These indicators are now increasingly used for monitoring and guiding the European forestry sector. However, even after 20 years of work, threatened species and populations—a central component of the basic sustainability concern—still remain poorly reported and considered (San-Miguel-Ayanz et al. 2011). One reason is that threatened biodiversity is too heterogeneous to be captured by a few broad indicators. Such a lesson from Europe calls for better linkages between forest management and traditional biodiversity conservation on a global scale.

Iconic species—conspicuous, well-known, and socially valued organisms, which are currently under threat—provide obvious opportunities for establishing the missing forestry–conservation links. In conservation, such species have been successfully used as 'flagships' for education, fund raising, and establishing reserves (Caro 2010), and they can be smoothly incorporated into ecosystem service accounts (e.g. Werner et al. 2014). Iconic species logically position at the forestry–conservation interface when they are threatened by such landscape-scale processes that cannot be mitigated without improving forestry practices. A relevant forest-management response could qualify under the 'focal species' framework, which assumes that managing a landscape for selected demanding species conserves wider biodiversity (Lambeck 1997). A distinct feature is that, due to their perceived value, iconic species are likely to be accepted as forest management foci simultaneously by foresters, forest owners, conservationists and other stakeholders (such as hunters or ecotourism operators). Yet, there are surprisingly few cases worldwide where the protection of iconic species has expanded to forest landscape management—large carnivores in Europe (Chapron et al. 2014), white-backed woodpecker (*Dendrocopos leucotos*) in Sweden (Roberge et al. 2008) and northern spotted owls (*Strix occidentalis caurina*) in the United States (USFWS 2011) are among the few explicitly considered. We see a great potential for iconic species in participatory forestry planning (see also Spies et al. 2007; Santangeli et al. 2013), notably for attracting the private sector—a much-hoped-for key player in the green economic transition (Gasparatos and Willis 2015).

Western Capercaillie (*Tetrao urogallus*) is a big, largely sedentary forest grouse that inhabits old conifer and mixed forests across the Palearctic. Its global population is not threatened but regional declines prevail in Europe and the species has become red-listed in many countries (Storch 2007). Thus its status over most of Europe has been shifting from a highly valued game bird to a protected species, as revealed by a mixed position in the EU Birds Directive (2009/147/EC): special protection measures (Annex I) combined with legal hunting in certain member states (Annex II). Intensive forest management is considered to be the main threat: it degrades and fragments Capercaillie's old-forest habitat, reduces the abundance of food plants, and supports predators (Sjöberg 1996; Storch 1997; Kurki et al. 2000). For that reason, the Capercaillie has been proposed as a conservation flagship for a range of forest settings: the Finnish taiga (Pakkala et al. 2003), Scottish pinewoods (Summers 2007), spruce-fir forests in the Alps and Carpathians (Suter et al. 2002; Mikoláš et al. 2015), and beech stands in the Pyrenees (Segura et al. 2014). The overarching concept is a necessity for regional landscape approaches due to the species' large spatial requirements (e.g., Rolstad and Wegge 1989a; Storch 1997; Graf et al. 2007), geographical variation in habitat use (e.g. Graf et al. 2006), and tolerance of only certain forms and intensities of timber harvesting (Romanov 1960; Savchenko 2009; Miettinen et al. 2009; Wegge and Rolstad 2011). Hence, the Capercaillie could become a focal species for forest landscape management—supporting the much-needed flexibility in sustainability approaches (cf. Sayer and Maginnis 2005).

In this paper, we contribute to the idea of regionally adjusted management for focal species by reviewing the significance of the Capercaillie in the Baltic forest governance. The three adjacent Baltic States (Estonia, Latvia, Lithuania) encompass a total of 175,000 km^2 land area, including 71,000 km^2 of forest land. These countries share a distinct combination of natural, historical and land-use factors, which is promoting common initiatives for forest conservation and sustainable management (e.g., Kurlavičius et al. 2004; Lazdinis et al. 2009). The area is characterized by European mixed hemiboreal forests on flat or hilly low-lying terrains (maximum altitude 318 m a.s.l.), with a transition to the north-temperate deciduous forest zone in Lithuania (Ahti et al. 1968). Post-glacial development of wetlands has produced, inter alia, large raised bogs with their surrounding Scots pine (*Pinus sylvestris*) forests that form important Capercaillie habitat. In all three Baltic States, the history of forestry governance has much in common: (i) Russian and German roots, which developed to national forestry schools during the first republican period (1918–1940); (ii) similar centralized state-ownership and forestry planning during the Soviet occupation (1940–1991); (iii) parallel processes of re-privatization, turning to market economy, and accession to the European Union since the 1990s.

The Baltic Capercaillie populations were historically praised for their abundance and importance for hunters (Fischer 1791; Tyzenhauz 1842–1846; Martenson 1899) but reports on local declines appeared in the second half of the 19th century (e.g., Nolcken 1870). During the 20th century, the once extensive breeding range of the species significantly contracted and the populations declined in all three countries (e.g., Ivanauskas 1957; Viht 1974; see below for an update). Many factors have been blamed for those declines: uncontrolled hunting and poaching, timber harvesting, grazing, wetland drainage, predation, and climate change; yet, forestry (notably clear-cutting and drainage) has always been a central concern (e.g., Mihelsons 1958; Viht and Randla 2002). Within a few decades in the 20th century, the Baltic Capercaillie hunting tradition dwindled to strictly controlled shooting of single cocks from selected leks, and game management gave way to habitat protection. (Still, the species remains the symbol of the national hunters' society in Estonia.) By the 21st century, the conservation efforts in every country have further intensified—to include national action planning and a status of the most protected forest species by area (Table 1).

Despite such long-term attention on the Capercaillie and obvious similarities across Baltic countries, the knowledge base and lessons learned have not been integrated and communicated to the research community. In the current work, we fill this specific gap by synthesizing recent surveys and historical data into a broad overview on the Baltic Capercaillie's decline, current status, and management perspectives. In the first part of our review, we analyse changes in Capercaillie distribution, abundance, and critical habitats based on large-scale census of lekking males in all three countries. In the second part of our review, we describe how its population reductions have motivated a flow of innovative approaches to conservation management, which have now reached a phase of actively combining different management approaches through stakeholder communication. The significance of such historical perspective is a demonstration of a disproportionately large, long-term influence that an iconic species can have on developing sustainable land management.

Table 1 Current numbers and protection of Baltic Capercaillie leks

Variable	Estonia	Latvia	Lithuania	Total
Lek census (last count in each lek)				
A. Total no. of leks	413	454	86 (94)[a]	953 (961)[a]
B. Estimated no. of males at leks	1243	1932	212 (275)[a]	3387 (3450)[a]
C. Well-surveyed leks	403	375	76	854
D. Minimum no. of males in C	1163	1069	151	2383
E. Mean estimated lek size (B/A)	3.0 ± 2.1	4.3 ± 2.6	2.5 ± 1.2	3.6 ± 2.4
F. Mean minimum lek size (D/C)	2.9 ± 2.0	2.9 ± 1.9	2.0 ± 1.1	2.8 ± 1.9
Lek protection				
No. of leks in general reserves (%)	211 (51)	111 (24)	40 (42)	362 (38)
No. of specially protected leks (%)	160 (39)	299 (66)	46 (49)[b]	505 (53)
No. of leks without protection (%)	42 (9)	44 (10)	8 (9)	94 (10)
Area specially protected, ha	60,342	45,159[c]	11,785	117,286

Variable C refers to a subset of leks with the last count maximum 5 years old and variable D is the number of actually lekking males observed in these leks. For variables A and B, pre-2010 counts have been incorporated for the leks without more recent data, and the population estimate refers to all adult males encountered at the leks. The lek sizes represent mean ± SD

[a] The numbers in brackets include eight Lithuanian leks with only rough size estimates available (of these, six leks with 55–60 males in 2014–2015 inhabit the Čepkeliai reserve; E. Drobelis and E. Klimavičius, pers. comm.). *Lek protection* refers to those leks that were inhabited at the last count, except the total area (ha), which includes some sites where the lek is known to have been extirpated recently

[b] Special Protected Areas designated for the Capercaillie according to the EU Birds Directive

[c] Additionally, 47 leks are voluntarily protected by LVM according to the procedure described in the Online Resource 2

Material and the analytical approach

The central unit of observation in our review is a Capercaillie lek, which constitutes a spatial aggregation of spring territories of adult males and the place of courtship. Capercaillie leks have long fascinated naturalists and gamekeepers, resulting in some of the most detailed historical records available for any wild population (e.g., Suchant and Braunisch 2004). Interpreting lek counts in terms of population size is not straightforward due to variable sex ratio in Capercaillie populations (Wegge and Rolstad 2011; Mollet et al. 2015), but repeated broad-scale census can provide reliable information on distribution range dynamics (e.g., Mollet et al. 2003). Studies on the Greater sage-grouse (*Centrocercus urophasianus*) indicate that extensive lek count data are generally robust for establishing population trends in tetraonids, particularly in long time series (Fedy and Aldridge 2011; Blomberg et al. 2013; Monroe et al. 2016).

In all three countries, there exist national databases with Capercaillie lek locations that have been used for habitat protection purposes at least since the 1980s. These data have been recently updated with national lek inventories: in 2008–2012 in Lithuania (Zizas 2015), in 2009–2012 in Estonia (by the Estonian Ornithological Society; e.g., Leivits 2014), and in 2012–2014 in Latvia (archived at the JSC "Latvian State Forests"). The data have been updated by annual monitoring thereafter. The basic method has been a combined mid-spring (peak-season) census of males arriving to the lek site in the evening and present in the lek in early morning. Potential sites to visit have been identified by a wide range of methods, including signs of the birds' presence, suitable habitat, local information and historical records. The mapped positions of males have been used for delineating lek borders and its geometric centre. Single lekking males have been considered as 'leks' only when their full-season presence >1 km away of larger leks has been established. Leks in new locations have been carefully compared with the status of all previous data from the surroundings; if any other lek within 1 km had disappeared by the same season, the new locations were technically considered lek 'movements' and only the latest data were used for the analyses. For this overview, we have critically reviewed and harmonized these national databases to describe the current status of the Baltic Capercaillie population.

To quantify population change, we supplemented the recent survey data with historical counts. We performed thorough searches and critical assessment of all available literature (e.g. Ivanauskas 1957; Logminas 1960, 1962; Rootsmäe and Rootsmäe 1983; Priednieks et al. 1989; Renno 1993; see also Online Resource 1), the archives of national ornithological societies (comprising local reports), and the personal archives of Prof. E. Kumari (deposited at the Estonian University of Life Sciences) and E. Viht in Estonia. Based on these, we could identify lek areas that have disappeared and to assess population fragmentation.

For a robust assessment of population trends and critical events, we supplemented the distribution analysis with three complementary approaches based on lek counts. Raw counts are sensitive to timing, observer experience, inclusion and detection of passive males, and the attendance of younger males in the lek (Romanov 1988; Jacob et al. 2010). This can explain why historical population estimates of Baltic Capercaillie have varied two-fold and more: for example, ca. 3500 versus up to 10,000 adult males in Estonia in the 1940s (Kumari 1954; Viht 1974) and 4875 versus 8000 adult birds in Latvia in 1926 (Kalniņš 1943; Priednieks et al. 1989). Another problem is converting lek numbers to bird numbers, for example, interpreting the 30 % loss of known leks in Estonia in 1970–2000 (Viht and Randla 2002). In this paper we (i) searched the archives for comprehensive historical surveys in large plots to be compared with the current surveys, to reach estimates that are not influenced by lek movement (cf. Suchant and Braunisch 2004); (ii) quantified the mean size of repeatedly counted leks; and (iii) followed the trends in maximum lek sizes reported through the history. To analyse trends in the mean lek size in the large Estonian dataset (a total of 1601 male counts at 445 leks) we used generalized mixed additive modelling (Poisson distribution; log link function) with gamm4 package in program R (Wood and Scheipl 2014). Lek identity was included as a random factor and non-linearities were smoothed by univariate penalized cubic regression splines (df = 7.8).

To illustrate the habitats and habitat changes that are in the centre of our discussion on conservation strategies, we used the surveyed lek-centre positions in GIS analyses at two scales. At the landscape scale (within 3-km radius), we measured land cover proportions from the 1:100,000 Corine Land Cover 2006 vector map, and we also calculated the Shannon index of land-cover heterogeneity based on the 3rd order cover types (EEA 2007). In the same radius, we estimated changes in forest cover according to the data from Hansen et al. (2013); we present these relative to Corine 2006 forest areas. Finally, we checked for the link between landscape factors and lek size (an indicator of population change). For that we used the minimum number of males at the last count as the dependent variable in another generalized linear model (Poisson distribution; logit link function; country included as a covariate to account for possible census differences).

The stand-scale descriptions (within 250-m radius) include data on site type, forest composition, age, and timber volumes extracted from official forestry databases (based on full-area field surveys). We harmonized the national site typologies, using Bušs (1997) as a basis for dry and fresh site types, and Lõhmus (1984) for paludified and peatland site types. In total, we distinguished 21 forest site types (drained and undrained paludified/peatland forests treated as separate "types"). We also illustrate latitudinal variation in these characteristics, using simple linear regression with latitude (°) as the independent variable.

The declining population

Contraction of the distribution range

The Baltic Capercaillie leks have been surveyed for more than a century. Initially, such surveys were motivated by the land-ownership based hunting tradition that valued spring hunting from leks; later in the 20th century the interest transformed to lek-centred conservation approaches (see below). Between 1926 and 1940, the Estonian Capercaillie hunting scheme on state lands derived hunting quota directly from early-spring counts. At its maximum in 1938, for example, a total of 2435 cocks were reported in 475 leks, after which the State Forest Service issued 425 shooting permits and 291 cocks were actually hunted (Teino 1939). Such individual accounts were gradually integrated into population perspectives: the first national distribution map (depicting Capercaillie densities by forest districts) was prepared by Kalniņš (1943) in Latvia, while a multi-scale conservation approach on the lekking population was developed by Logminas (1960, 1962) in Lithuania. Although late-summer transect counts were introduced in 1978 in Estonia and in 2012 in Latvia, the overall low population density (<0.2 adult birds per km^2 of forest) and its patchy, lek-centred distribution pattern have so far limited the added value of transect counts to rough estimates of reproductive success (Viht 1990). Thus, leks remain the main census targets, but the methods have evolved and past estimates require critical appraisal.

As of 2015, we can confirm the existence and location of 961 contemporary Capercaillie leks with an estimated 3450 lekking males in the Baltic States (Table 1). This population is split between six larger areas (ca. 130–1100 km^2) and about twenty small fragments (Fig. 1). In general, Lithuanian leks are the smallest, while the apparently larger size of

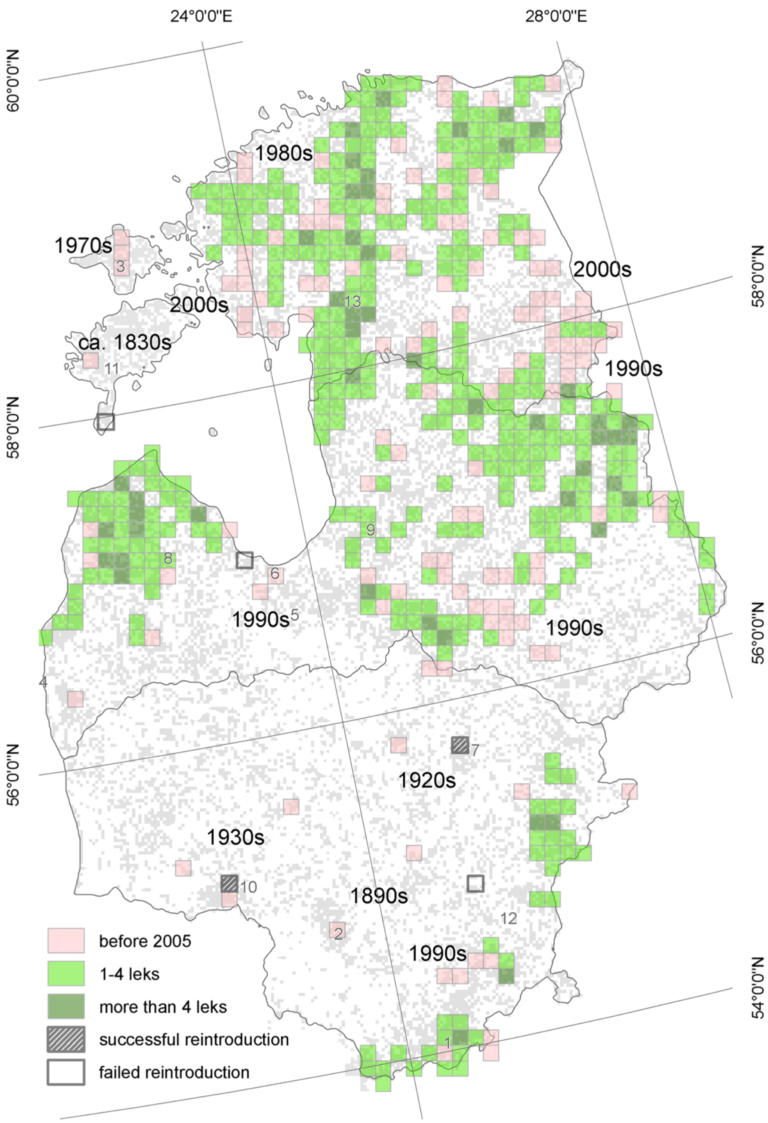

Fig. 1 Historical and current distribution of Capercaillie leks in the Baltic States (ETRS LAEA 5210 10 × 10 km grid). Extant leks (*green* cells) refer to the period 2006–2015 based on national surveys; extirpations (*red* cells) are depicted only for documented historical leks. Decades indicate the approximate time of regional extirpation. The locations mentioned in the text are depicted with numbers: *1* Čepkeliai; *2* Kazlų Rūda–Zapyškis forest and peatland area; *3* Hiiumaa island; *4* Liepāja; *5* Jelgava; *6* Jurmala; *7* Šimoniu forest; *8* Stende; *9* Allaži; *10* Karšuva forest; *11* Saaremaa island; *12* Vilnius; *13* Soomaa. The *grey* background pattern shows forest distribution according to the Corine Land Cover 2006 (EEA 2007)

Latvian compared to Estonian leks may be related to estimation techniques (Table 1; note the similarity of the strict estimate 'F'). The largest well-surveyed lek is situated in northeastern Latvia (18–20 males); the top counts for Estonia and Lithuania being 15–18 and 5–6 males, respectively. However, there is an observation of an aggregation of ca. 30 males in a remote 100-ha area in the Čepkeliai reserve in Lithuania (E. Drobelis and E. Klimavičius, pers. comm.), which suggests that the northward increase in mean lek size is due to current habitat quality rather than geography per se.

Disappearance of documented leks from 26 % of formerly inhabited 10 × 10 km squares alone (Fig. 1: red squares) indicates that the range has severely contracted during the last century. The true reduction has been probably steeper, particularly in the south, as judged by historical maps (Kalniņš 1943; Priednieks et al. 1989) and accounts on regional extirpations (e.g., Ivanauskas 1957; Viht 1974; Viht and Randla 2002). The range contraction (Fig. 1) has included three main processes:

- Population remnants in the *southern agricultural region* were lost more than 80 years ago, creating a large distribution gap. A 19th century key event was the railway construction through the Kazlų Rūda–Zapyškis forest and peatland area. This economically important connection between Russia and Poland opened the last South-Baltic wildland to exploitation and poaching, and isolated the south-western populations, which were eventually lost in the 1920s due to animal grazing and unregulated hunting (Ivanauskas 1957);
- Both *island populations* have been lost, the better documented and more recent being the 70-year decline to extirpation of the Hiiumaa population that comprised ca. 100 lekking cocks in the early 1900s (Loudon 1909);
- Strongholds of the species in *mainland Estonia and Latvia* have been shrinking around high-density distribution centres and becoming increasingly fragmented. The West-Latvian (Courland) population was isolated already in the 1930s (Kalniņš 1943) but the gap has deepened with the loss of small populations in Liepāja and Jelgava regions. However, only within the last 20 years has the main northern population been split into three large parts—South-Latvian (Zemgale) population, the main Latvian population (together with South-Estonian), and the main Estonian population (including some leks in extreme northwest Latvia) (Fig. 1). These recent events suggest that the whole population is prone to further contraction and isolation.

There have been at least five attempts of Capercaillie reintroduction in the Baltic States. In 1911–1912, a total of nine males and 14 females were reintroduced to Saaremaa Island in Estonia; in spring 1912 a few cocks were observed displaying and one brood fledged in 1914 but the birds then disappeared (Toll 1987). Between 1966 and 1979, there were a total of six releases—in total, 79 males and 99 females originating from Kirov region (Russia)—to Latvia and Lithuania (Chlevickas 1978; Romanov 1988). The birds were of non-native subspecies (*T. urogallus uralensis/pleskei*, instead of the native *T. urogallus major*) and the reintroductions had limited success. The attempt near Vilnius in 1974 reportedly failed due to "wrong habitat"; the birds released near Jurmala in 1966 initially reproduced but the whole population disappeared by 2001. Only the double reintroduction (1973, 1979) in Šimoniu forest in northeastern Lithuania resulted in two leks by the 1980s, one of which survives until today. Two other releases of birds from Kirov (in total, 11 males and 20 females) augmented small populations near Stende in western Latvia and Allaži in central Latvia in 1969; their impacts are unknown but both these populations currently exist.

The only modern reintroduction, based on native subspecies, soft-release techniques and habitat management, is ongoing in Karšuva forest in southern Lithuania. Eggs are being incubated and chicks reared in a special predator-proof vivarium on a forest edge at the release site until October. Habitat management includes the removal of spruce undergrowth and no-kill trapping of mammalian predators, such as *Martes martes, Vulpes vulpes* and the invasive *Nyctereutes procyonoides*, in ca. 10-km radius. In 2011–2014, the first 23 individuals (7 males, 16 females) have been released. Most of them settled within a few kilometers (maximum 21 km) from the release place (Butleris 2014; A. Butleris, pers. comm).

Extent and timing of the population decline

Data on multiple large plots with historical data from current Capercaillie range were only found for Estonia. Based on four plots totalling 1020 km^2, the current densities of lekking males form 29 $\pm$ 8 % (95 % CI) of mid-20th century levels (Online Resource 1). A broad pattern of population decline was, however, supported by consistent negative trends in all countries for the mean size of repeatedly counted leks and the occurrence of very large leks. The mean lek size declined from 8.4 to 2.9 males in 15 surviving Lithuanian leks that were censused in 1953–1954 (Logminas 1960) and in 2008–2012, and from 4.3 to 2.8 males just between 2000 and 2009 in a sample of 28 Latvian leks. Explicit analysis of the large Estonian dataset distinguishes lek reductions in the 1970s and the 2000s (Fig. 2; also Viht 1988), while earlier population declines typically involved disappearances of whole leks (Viht 1985). The largest Baltic leks were once famous across Europe; for example, the 19th century counts could reach 50–60 lekking males and twice as many attending the lek in historical Livonia (Krüdener 1898). The last lek of such size was still known in 1952 in North-Estonia and many leks with 20–30 males existed in the 1930s and 1940s (Kask 1973; Online Resource 1). Currently, only three Baltic leks host more than 15 males (one in each country, see above).

In addition to the population decline effect, the lek-size changes reported have apparently included some fragmentation of leks on the landscape since the appearance of small leks (1–3 males) was noticed as a new phenomenon by naturalists at least in Estonia (Kask 1973). Studies in other parts of the range have indicated that such restructuring of Capercaillie leks is caused by habitat loss and landscape fragmentation (Rolstad and Wegge 1989b; Rolstad et al. 2009). Since such fragmentation is more likely for large leks

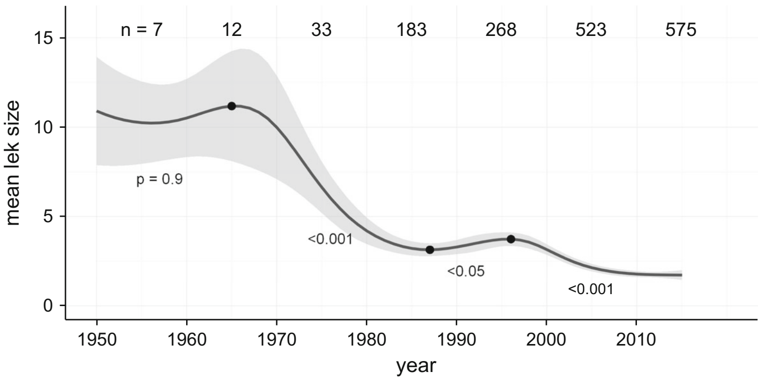

Fig. 2 The mean size of repeatedly censused Capercaillie leks in Estonia (95 % confidence intervals as the *grey area*), 1950–2015. *P*-levels refer to trends between the turning points of the *line*; number of observations by decade is shown on *top*

that are also easier to record, the 1970s decline in the total number of lekking males in Estonia may be less steep than Fig. 2 suggests.

Importantly, the overall population contraction and decline has included periods of stability and local recoveries. In Estonia and Latvia, there are multiple consistent reports on significant increases (unexplainable by improved surveys alone) in the 1920s−1930s, following the massive decline during and after the World War I (Kalniņš 1943; Viht 1974). In Estonia, that recovery was mostly explained by strict hunting rules (including regional bans) and control of poaching since 1926 (Teino 1939) and there were probably at least 6500 lekking males after that (estimated by combining current numbers, loss of the national range and the 60-year reduction within the range: Table 1, Fig. 1, Online Resource 1). Next, after the 1970s crashes, the 1980s stand out as relatively favourable in all three countries—as supported by official lek counts and (in Estonia) stability or slight increase both in the mean lek size (Fig. 2) and late-summer transect counts (Viht 1990). In Estonia and Lithuania, such trend has been maintained later in some large protected areas only: Lahemaa and Alam-Pedja (Online Resource 1) and Čepkeliai (40–50 lekking males in 1980–2004, 55–60 males in 2014–2015; E. Drobelis and E. Klimavičius, pers. comm.). The stability may have been more widespread in Latvia, as judged by stable official counts and largest-lek size.

The focal habitat

Baltic Capercaillie inhabit landscapes dominated by coniferous and mixed forest and transitional woodland, such as wooded mires (Table 2). These land cover types formed a total of 83 ± 13 % (SD) within 3 km around 836 extant leks, varying little between the countries (82 % in Estonia and Latvia, 90 % in Lithuania). However, the landscapes are naturally more heterogeneous towards north, due to a decrease in forest cover (particularly coniferous) and an increase of transitional woodlands (also open wetlands; Table 2). These typical land cover features (except landscape heterogeneity) also support larger leks. However, the low explanatory power of such broad-scale model (Table 3; note also the longitudinal effect, i.e., smaller leks in the peripheral western areas) suggests that lek size is mostly determined by smaller-scale factors.

Baltic Capercaillie landscapes have recently lost forest cover, mostly through intensified logging (Table 2). The mean net losses compared to the reference year 2006 are 2.8 ± 0.7 % in Lithuania, 3.8 ± 0.4 % in Estonia and 5.1 ± 0.5 % in Latvia. However, forest loss is unrelated to lek size (Table 3), probably because concentrated clearcutting usually eradicates whole leks (Viht 1985; Viht and Randla 2002). Hence, the lek-size reductions (see above) should have another explanation, such as increased predation pressure or deterioration of existing habitat.

The leks are typically situated in old pine-dominated woodland. Most stable features are a high proportion of Scots pine in the tree layer (81 ± 1 % within 250 m; N = 738 leks) and predominance of acidic soils, while the actual site type composition and stand structure vary regionally. Forests on mineral soil dominate in Lithuania, notably Myrtillosa (bilberry) site type, but towards north lek sites are increasingly in forested wetlands and semi-open mires (Fig. 3). This creates a gradient of lek protection costs: despite an increase in forest age (Table 2), the mean timber volumes growing around the leks decrease twofold from southern Lithuania to northern Estonia (Fig. 4). Compared to the northward trends of the landscapes, wetlands become even more significant at the lek-site scale and, along with landscape heterogeneity, lek sites show an increased diversity of forest types (Fig. 3; Table 2).

Table 2 Landscape and forest features around Baltic Capercaillie leks (occupied 2006–2014)

Characteristic	Mean ± SD (Country average)			Northward trend	
	Estonia	Latvia	Lithuania	Mean slope ± SE	*P*
Landscapes within 3 km radius					
Land cover % (CLC code)					
Forests (3.1)	64 ± 13 (48)	65 ± 15 (41)	83 ± 10 (29)	−3.2 ± 0.4	<0.001
Conifer forest (312)	35 ± 17 (18)	37 ± 17 (14)	64 ± 16 (11)	−5.5 ± 0.4	<0.001
Mixed forest (313)	23 ± 12 (19)	22 ± 12 (18)	17 ± 10 (11)	−1.5 ± 0.3	<0.001
Transitional woodland–shrub (324)	18 ± 8 (10)	17 ± 8 (10)	7 ± 7 (4)	2.0 ± 0.2	<0.001
Inland wetlands (4.1)	8 ± 11 (5)	5 ± 8 (2)	3 ± 5 (1)	1.4 ± 0.2	<0.001
Agricultural areas (2)	9 ± 9 (34)	13 ± 12 (45)	7 ± 7 (63)	0.2 ± 0.3	0.539
Artificial surfaces (1)	<1 (2)	<1 (1)	<1 (3)	0	0.678
Other land-cover statistics					
Shannon diversity of land cover	1.49 ± 0.25	1.45 ± 0.28	1.04 ± 0.38	0.09 ± 0.01	<0.001
% core habitat of conifer forest	41 ± 13	41 ± 14	61 ± 12	0	0.943
Forest loss in 2000–2013 (%)	6 ± 6	10 ± 7	4 ± 4	0.3 ± 0.2	0.066
Forest gain in 2000–2013 (%)	2 ± 3	4 ± 3	2 ± 2	0.2 ± 0.0	<0.001
N	390	370	76	836	
Lek sites within 250 m radius					
Shannon diversity of site types	1.01 ± 0.47	1.07 ± 0.47	0.58 ± 0.42	0.06 ± 0.01	<0.001
Scots pine (%) (1st layer)	82 ± 19	79 ± 17	88 ± 16	−0.9 ± 0.5	0.086
Stand age, years	95 ± 24	93 ± 20	80 ± 17	3.1 ± 0.6	<0.001
Timber, m^3 ha^{-1} (forest)	177 ± 54	222 ± 52	276 ± 80	−22.9 ± 1.7	<0.001
Timber, m^3 ha^{-1} (all area)*	155 ± 68	198 ± 66	272 ± 80	−26.5 ± 2.1	<0.001
N	319 (*323)	354 (*355)	65	738 (*746)	

Northward trends have been estimated using linear regression, with latitude (°) as the independent variable. *Landscape sample* 836 leks having within 3 km >90 % area on the land territory of the Baltic States. Land cover refers to the 1:100,000 Corine Land Cover 2006 map (EEA 2007); all proportions are from land area. Core area omits 100 m of edge to adjacent cover types. *Lek site sample* 746 leks having forestry data for >90 % area within 250 m including, for forest structure calculations, ≥1 ha forest ($N = 738$)

The variable and samples including non-forest areas are marked with asterisk

Drainage is the main current human impact on lek sites, particularly in the north. Among all paludified and peatland forest areas around leks, 59 % are classified as "drained" in forestry terms in Estonia, 40 % in Latvia and 7 % in Lithuania. Clear-cuts form only minor areas: 10 % of extant leks have >4 % clear-cut area within 250 m. Pine trees in the lek sites have a relatively even age structure, although the variation can be underestimated particularly on peatlands. For example, in the Estonian stands with pine presence (a total of 9619 ha explored within 250 m of lek sites), only 11 % had at least two age-classes of pines, and only 7 % of lek sites had at least 5 % pine volume present in the second layer. The age of pines does not differ systematically from the rest of the overstorey.

Table 3 Broad-scale effects on Capercaillie lek size in the Baltic States (linear model for the minimum count of males in 836 well-surveyed leks, 2010–2015; median year 2013)

Factor	Regression coefficient ± SE	*P*
% conifer forest	0.005 ± 0.002	0.012
% transitional woodland-shrub	0.007 ± 0.003	0.028
% agricultural areas	−0.010 ± 0.002	<0.001
Shannon diversity of land cover	0.196 ± 0.142	0.165
Net forest loss in 2000–2013 (%)	0.002 ± 0.005	0.608
Longitude (°)	0.047 ± 0.014	<0.001
Model intercept	0.753 ± 0.457	0.100

The landscape factors correspond to those given in Table 2; *P*-values refer to likelihood-ratio tests; model pseudo-$R^2 = 0.06$

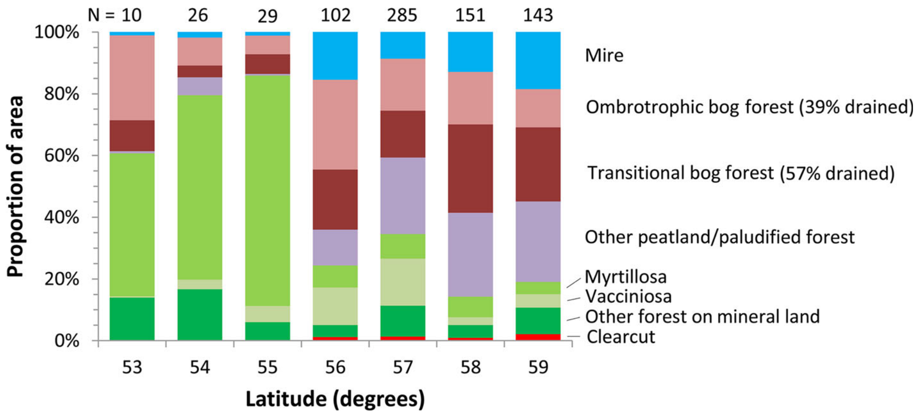

Fig. 3 Mean habitat composition within 250 m around Capercaillie leks (N = 746) along the latitudinal gradient in the Baltic States

Towards a focal-species approach

The management dimension

In northern Europe, Capercaillie is one of the best known forest species for forestry stakeholders, including forest owners (Uliczka et al. 2004). In addition to such passive recognition, this bird has long been at the forefront of innovative management approaches in Baltic forest governance (Fig. 5). Judging by the reviewed and reconstructed population trends, some of these approaches appeared effective in the short term. However, they all have proven to be insufficient in the long term—along with the emergence of new threats.

The first response to the 19th century declines was an expansion of *general predator control*, as applied regionally already in the 18th century (Fischer 1791). Predator control became a mainstream management approach particularly on the hunting lands of the Baltic-German noble class in Latvia and Estonia. Promoted by various orders, campaigns, and awards, the local elimination of mid-sized carnivores, raptors and larger corvids was highly effective and probably supported the huge Capercaillie leks of that time. The

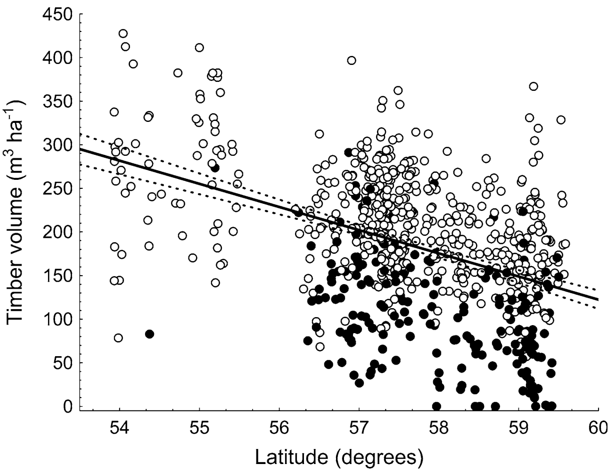

Fig. 4 Mean timber volumes within 250 m around Capercaillie leks (N = 746) in relation to the latitude in the Baltic States. The *regression line*, its 95 % confidence intervals, and the sites having >10 % non-forested area (mire; *filled symbols*) are shown. In multiple linear regression, latitude and mire proportion together explain 54 % of variation in the timber volumes observed

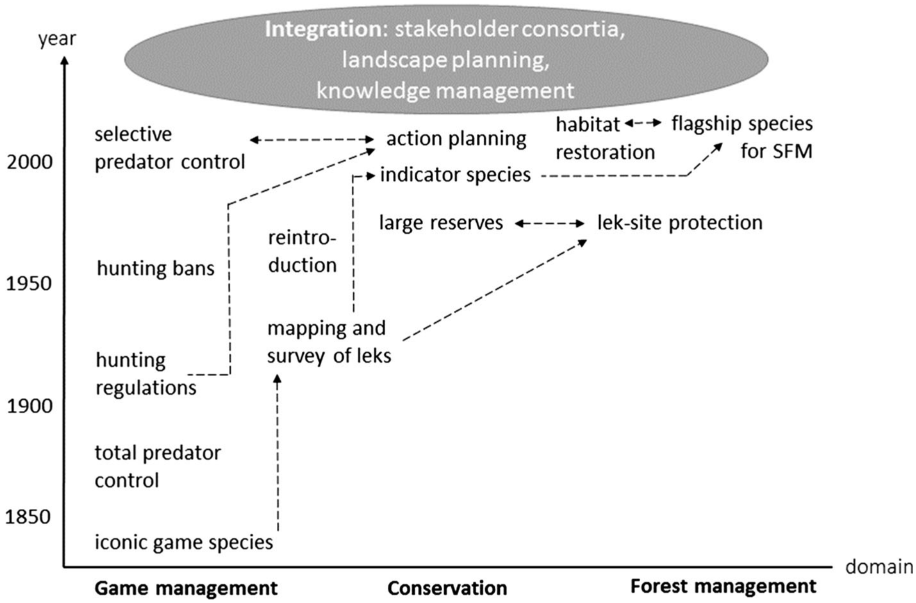

Fig. 5 Development of management approaches to the Baltic Capercaillie population in the three domains of activity and knowledge. Major inter-domain links have been identified by *dashed lines*. For example, its iconic status for hunters promoted comprehensive surveys of Capercaillie leks, which were later used in the lek-site protection approach by foresters and forest indicator approach in the conservation domain; these in turn provided a basis for its current flagship status in sustainable forest management (SFM)

number of killed raptors in Estonia alone reached 400,000 individuals until the 1960s (Lõhmus 2011). Predator control impacts were, however, explicitly reported only in the 20th century when they were already mixed with other management impacts (e.g. Kuku

1940). By that time, the persecution had also raised predator conservation concern and broad-scale culling was gradually prohibited.

During the World War I and time of reorganization of the young Baltic republics after 1918, poaching and unregulated hunting lead to massive declines in all game and local extirpation of some, including the Capercaillie. That issue was addressed in the mid-1920s by *strict national hunting regulations*, which (combined with local hunting sanctuaries and a revival of predator control) enabled partial recoveries in Latvia and Estonia (see above). Hunting of Capercaillie females has been prohibited in Estonia since 1934. In Lithuania, the same process and similar impacts were observed in the 1950s (Dautartas and Paltanavičius 1988). Some locations emptied by hunting also became the first targets of *reintroduction and augmentation*, which has been a learning process from early failures to modern techniques, and represents the only such experience for Baltic forest birds. Meanwhile, deterioration of major populations since the 1950s (e.g., Mihelsons 1958) increasingly revealed environmental threats that overrode population supplementation. These new threats gradually removed Capercaillie from game lists. The whole Baltic hunting bag totalled more than 1000 individuals annually in the 1930s, 200–300 cocks in the early 1970s, and fewer than 100 cocks (mostly in Latvia) since the 1980s (Teino 1939; Kalniņš 1943; Dautartas and Paltanavičius 1988; official statistics).

By the 1970s, Capercaillie habitat degradation due to clearcutting and forest drainage became so evident that it initiated special *lek protection* initiatives. There were two main approaches, which have contributed rather equally to establishing an unprecedented level of protection for any widespread species—now exceeding 90 % of known leks (Table 1). A broad approach among the conservation community was to use lek aggregations as a major justification for *large wilderness reserves*—pioneered by the Čepkeliai nature reserve establishment in Lithuania in 1975. Simultaneously, a *lek-centred* ('micro-reserve') approach emerged in forestry and game management sectors, typically excluding ca. 25–100 ha areas from commercial forest management. Initially, the protection was given to selected, "most valuable" leks only. Thus, a 1980 state regulation in Estonia permitted management restrictions (within 300 m of the lek) to "no more than three leks per 10,000 hectares of forest land". However, by the end of the 20th century, Capercaillie leks became established as the regionally most influential species-level indicator for identifying high-conservation-value forests (Kurlavičius et al. 2004).

Despite initial stability or short-term recoveries in protected sites (e.g., Dautartas and Paltanavičius 1988), Capercaillie range contractions and population declines have recently continued on a broad scale (Figs. 1, 2). The mitigation role of the near-complete lek protection is yet to be quantified but, obviously, it has not sustained the population. The possible reasons are delayed or broad-scale impacts, such as predation, habitat fragmentation, drainage, or access via roads. Consequently, the 21st century management response has been a movement toward knowledge-based ecosystem management, comprising both integrative and specific activities based on national action planning (Fig. 5). The underlying assumption is that the species can be sustained when areas around leks retain sufficient areas of old, open pine forest with abundant cover of dwarf shrubs and safe nesting sites; however, the quantitative targets and relative importance of different factors are only based on expert opinion (see the next chapter). Latvia has enforced the most elaborate prescriptions to integrate silviculture with the conservation of Capercaillie lekking areas, and pioneered in habitat restoration in micro-reserves. The *silvicultural prescriptions* for Latvian state forests include special zoning, seasonal restrictions and prescriptions for overstorey and undergrowth retention, thinning intensity, clear-cut size, regeneration type, and removal of felling residues (Online Resource 2). By 2015, the *restoration activities* had involved 711 ha managed by various thinnings (>90 % in young stands or for removing spruce undergrowth) and 7 ha of experimental restoration of forested wetland (Fig. 6, photos). These approaches are being adopted by the other countries, with specific additions, such as *selective predator control* in lek areas (two local initiatives of Pine Marten removal in Estonia) and the *integrated reintroduction* approaches in Karšuva, Lithuania (see above). All these approaches continue to be pioneering attempts of their kind in the Baltic forest governance.

Social and research dimension: the Estonian case

Large conservation expenditures that do not halt biodiversity decline are a source of social tension. In Estonia, a breaking point arrived in 2010–2012 when the forestry sector decisively turned down all updates to the Capercaille action plan, referring to insufficient evidence for stricter lek protection. Site protection was also opposed by game managers who advocated for predator management. This confrontation revealed knowledge fragmentation and lack of trust among stakeholders: while they all agreed that the knowledge was inappropriately put into practice (i.e., there were ‘knowing-doing gaps’ sensu Habel et al. 2013), each stakeholder built its argument around a (trusted) part of the information only. Ironically, the former action plan had prescribed key research (Viht and Randla 2002), which had not been implemented. These features indicate a deviation from problem-solving approach in policy-making: lack of interest in and ineffective transfer of knowledge, power issues, and, possibly, inability to integrate and express different forms of knowledge (Pregernig 2014). Similar failures are probably widespread in forest governance but species with large spatial requirements and complex ecologies may be particularly vulnerable. It has been shown for other species and ecosystems that knowledge fragmentation itself (notably when backed by premature publications in top journals) can create conservation failures despite intensive research (Gross 2005; Hutchings and Marco 2009).

The Estonian confrontation was resolved by introducing an academic process of *shared knowledge production* (Fig. 5). Stakeholder collaboration is increasingly advocated in nature conservation based on general communication theories (e.g., Jones-Walters and Çil 2011) and on evidence of its social benefits that can indirectly improve conservation outcomes (Young et al. 2013). Our effort was specifically focused on increasing both the quality of the knowledge base (see MacMillan and Marshall 2006) and on trust-building among stakeholders (e.g., Gulbrandsen 2008; Young et al. 2013). In 2012, a ‘think-tank’ with representatives of all the stakeholders met for a collective effort: to identify and prioritize knowledge gaps for Capercaillie conservation in Estonia on a consensus basis, and to agree on the procedures and responsibilities for addressing those gaps. The prioritization exercise required each research gap to be explicitly linked with its expected application in order to address potential ‘thematic gaps’ in researcher-manager communication (see Habel et al. 2013). The stakeholders listed a total of 20 research issues and formed a consortium to jointly address them. Five issues were highlighted as top priorities: (1) habitat use and disturbance sensitivity of adult birds outside the lekking season (telemetry study); (2) diet analyses of major predators (e.g., the Pine Marten and Raccoon dog) for assessing predation pressure; (3) experimental habitat management (stand structure; drainage impact) around small leks to understand habitat limitation and possibilities to restore impoverished populations; (4) documenting spatial dynamics of leks and their relationship with behavioural adaptation for long-term perspectives of the micro-reserve approach; (5) comparative spatial and retrospective population studies to discover long-

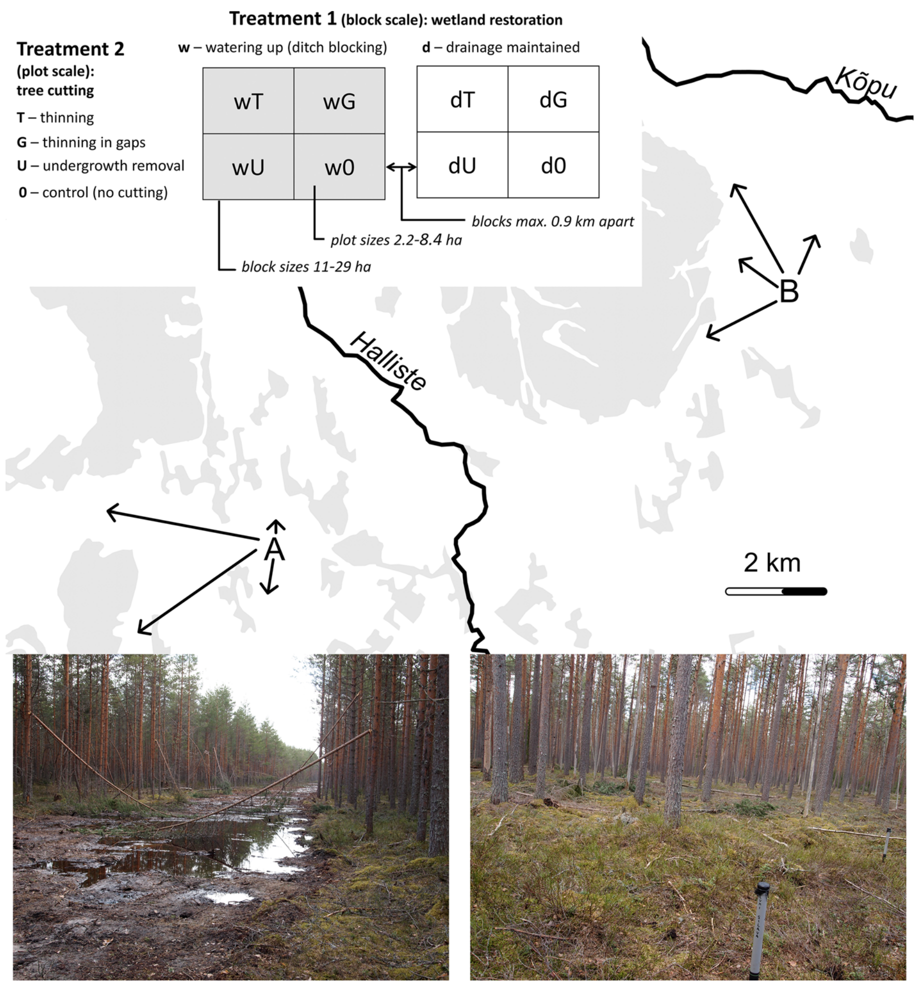

Fig. 6 Design of the Capercaillie habitat management experiment in Soomaa, southwestern Estonia. *Background map*: there are 4 + 4 experimental sites (pointed by *arrows*) 0.4–3.5 km away from two leks (*A*, *B*) in heavily drained forest landscapes near large mires (*grey areas*). *Treatment scheme* each experimental site comprises a pair of blocks, one of which undergoes wetland restoration by ditch filling. Each *block* is divided into four *plots*, which receive randomly assigned cutting treatments. Undergrowth is removed (retaining 20–30 dense spruces ha^{-1}) in all treatments except control; it is supplemented with thinning in treatments T (up to 30 % of volume harvested throughout the stand) and G (thinning in 40–100 m diameter gaps covering half of the forest area). *Photos* treatment "wT" in a test site in Latvia (filled ditch and adjacent thinned stand on 30 April 2013). Photo credits: A. Lõhmus

term, abrupt, and interacting threats for long-term conservation planning. The State Forest Management Centre agreed to provide initial financial support for the research, the Environmental Board to raise funds for a continuation, and the universities to adjust their research programs and to recruit graduate students.

In the next 3 years, the consortium (i) established an intensive study area in a regional core population, incorporating both commercially managed and protected landscapes; (ii) implemented a large replicated experiment in the centre of that area to test habitat management and restoration options; (iii) completed pre-surveys on Capercaillie nesting habitats and possible side-effects of the management; and (iv) analysed predator diets for grouse remains and equipped adult Capercaillie with GPS transmitters in the area. The experimental sites, which encompass ca. 3 km^2 in 16 paired blocks and a total of 64 plots, are situated in homogeneous drained pine-wetland landscapes at varying distances from two medium-sized leks (Fig. 6). The habitat interventions were implemented in 2015–2016 according to pioneering experience in selected Latvian sites (Fig. 6) and studies elsewhere—notably on thinning impacts on ground-level forage (Broome et al. 2014), importance of feeding trees and shade (e.g., Gjerde 1991), and site-dependent long-term perspectives (Braunisch and Suchant 2008). In addition to Capercaillie research, University of Tartu has been introducing general research of drained forest functioning into the same setup. By integrating basic and applied scientists, Capercaillie has thus been instrumental in closing another research-management gap—the 'disciplinary gap' (Habel et al. 2013).

A sequel to the 'think-tank' took place in 2015 where the stakeholders used the accumulated research results to plan a set of practical habitat improvement projects for next years. They consensually acknowledged that the consortium had proven useful as the central body for strategical planning of Capercaille conservation and established a rotational basis for consortium leadership. As of 2016, a similar research-based process is emerging in Latvia. We hope that these stakeholder processes help to avoid some communication mistakes that have hampered large-scale conservation efforts elsewhere, e.g., in the case of the White-backed woodpecker in Sweden (see Blicharska et al. 2014).

Conclusions

The merits and limitations of flagship species are well known regarding particular conservation situations (Caro 2010). Such issues are prominent also in the Baltic Capercaillie case: relatively large conservation expenditures both in land allocation and human effort, which might support forest biodiversity in general. An explicit understanding of the latter—cost-effectiveness in terms of broader ecosystem benefits—is still missing for this particular system and should be addressed in the future.

Our main conceptual contribution to this general framework is a historical perspective: we demonstrated how concern for an iconic species has supported long-term development of forest conservation approaches. Through the time and along with the emergence of new environmental risks, the Capercaillie has inspired innovative management ranging from specific techniques to sectoral integration. It is remarkable that such an active interest has been maintained throughout the multiple, and often dramatic, changes as experienced by the Baltic societies since the 19th century. Concern for Capercaille still brings together a roughly similar set of stakeholders (foresters; game managers; naturalists/conservationists; land owners), despite reorganizations in their approaches and the legal framework. Under such interest even stakeholder confrontation, as illustrated in the case of Estonia, can eventually increase the credibility, salience, and legitimacy of the knowledge system for sustainable forestry in general (see Cash et al. 2003). We thus see iconic species as a cultural stabilization mechanism for sustainable development.

The future of the Baltic Capercaillie population remains insecure, however. In Lithuania, its status may already be on the verge of critical isolation to require large-scale recovery programs similarly to the situation in the neighbouring Poland (Zawadzka 2004). The populations in Latvia and Estonia are much stronger, but our data show that these are becoming isolated, too. The reason cannot be simple habitat loss because forest cover has almost doubled in these countries during the last century. In Estonia, for example, the area of >100 year-old pine forests has also increased from ca. 45,000 ha in 1958 to more than 93,000 ha in 2012 (Keskkonnaagentuur 2014). It is therefore likely that Baltic Capercaillie management for the future requires another change in forest governance, which fortunately appears to be on the way.

Acknowledgments This paper is a tribute to hundreds of devoted bird-watchers, foresters, and gamekeepers across the Baltic States who have collected information on the locations and size of Capercaillie leks. We are particularly grateful to the following persons who contributed large datasets or consulted us during the manuscript preparation: Algis Butleris, Eugenijus Drobelis, Fred Jüssi, Evaldas Klimavičius, Tiit Randla, Vahur Sepp, Gytautas Stukas, Indrek Tammekänd, Tomas Tukačiauskas and Ene Viht. Eerik Leibak critically reviewed the data of the first Estonian bird atlas, and members of the Estonian Capercaillie consortium read the social description provided in the manuscript. Erik Blomberg, an anonymous reviewer and an associate editor provided helpful comments on the full draft. The national surveys in 2008–2015 were funded by the Estonian National Environmental Board and JVC "Latvia's State Forests". While compiling the paper, A.L was supported by the Estonian Research Council (Grant IUT 34-7).

References

Ahti T, Hämet-Ahti L, Jalas J (1968) Vegetation zones and their sections in northwestern Europe. Ann Bot Fenn 5:169–211

Blicharska M, Baxter PWJ, Mikusiński G (2014) Practical implementation of species' recovery plans—lessons from the White-backed Woodpecker Action Plan in Sweden. Ornis Fenn 91:108–128

Blomberg EJ, Sedinger JS, Nonne DV, Atamian MT (2013) Annual male lek attendance influences count-based population indices of greater sage-grouse. J Wildl Manag 77:1583–1592

Braunisch V, Suchant R (2008) Using ecological forest site mapping for long-term habitat suitability assessments in wildlife conservation—demonstrated for Capercaillie (*Tetrao urogallus*). For Ecol Manag 256:1209–1221

Broome A, Connolly T, Quine CP (2014) An evaluation of thinning to improve habitat for capercaillie (*Tetrao urogallus*). For Ecol Manag 314:94–103

Bušs K (1997) Forest ecosystem classification in Latvia. Proc Latvian Acad Sci B 51:204–218

Butleris A (2014) Reintroduction of Capercaillie in Karšuva forest. Paukščiai 12(4):18–21 **(in Lithuanian)**

Caro T (2010) Conservation by proxy: Indicator, umbrella, keystone, flagship and other surrogate species. Island Press, Washington, DC

Cash DW, Clark WC, Alcock F, Dickson NM, Eckley N, Guston DH, Jäger J, Mitchell RB (2003) Knowledge systems for sustainable development. Proc Natl Acad Sci USA 100:8086–8091

Chapron G, Kaczensky P, Linnell JDC, Von Arx M, Huber D, Andrén H, López-Bao JV, Adamec M et al (2014) Recovery of large carnivores in Europe's modern human-dominated landscapes. Science 346:1517–1519

Chlevickas J (1978) Capercaillie in Šimoniu forest. Mūsų gamta 2:5 **(in Lithuanian)**

Dautartas M, Paltanavičius S (1988) Status of the Capercaillie in Lithuania. In: Abstracts of the XII Eastern Baltic Ornithological Conference. Vilnius, pp 67

EEA (European Environment Agency) (2007) CLC2006 technical guidelines. EEA Technical report 17. European Environment Agency, Copenhagen

Fedy BC, Aldridge CL (2011) The importance of within-year repeated counts and the influence of scale on long-term monitoring of sage-grouse. J Wildl Manag 75:1022–1033

Fischer JB (1791) Versuch einer Naturgeschichte von Livland. Nicolovius, Königsberg

Gasparatos A, Willis KJ (2015) Biodiversity in the green economy. Routledge studies in ecological economics. Routledge, Oxon, p 40

Gjerde I (1991) Cues in winter habitat selection by Capercaillie. II. Experimental evidence. Ornis Scand 22:205–212

Graf RF, Bollmann K, Sachot S, Suter W, Bugmann H (2006) On the generality of habitat distribution models: A case study of Capercaillie in three Swiss regions. Ecography 29:319–328

Graf RF, Bollmann K, Bugmann H, Suter W (2007) Forest and landscape structure as predictors of Capercaillie occurrence. J Wildl Manag 71:356–365

Gross L (2005) Why not the best? How science failed the Florida panther. PLoS Biol 3:e333
Gulbrandsen LH (2008) The role of science in environmental governance: competing knowledge producers in Swedish and Norwegian forestry. Global Environ Politics 8:99–122
Habel JC, Gossner MM, Meyer ST, Eggermont H, Lens L, Dengler J, Weisser WW (2013) Mind the gaps when using science to address conservation concerns. Biodivers Conserv 22:2413–2427
Hansen MC, Potapov PV, Moore R, Hancher M, Turubanova SA, Tyukavina A, Thau D, Stehman SV, et al (2013) High-resolution global maps of 21st-century forest cover change. Science 342:850–853. (The database at http://earthenginepartners.appspot.com/science-2013-global-forest. Accessed on April 2015)
Hutchings JA, Marco FB (2009) Scientific advice on species at risk: a comparative analysis of status assessments of polar bear, *Ursus maritimus*. Environ Rev 17:45–51
Ivanauskas T (1957) Birds of Lithuania, vol 1. Valstybinės politinės ir mokslinės literatūros leidykla, Vilnius **(in Lithuanian)**
Jacob G, Debrunner R, Gugerli F, Schmid B, Bollmann K (2010) Field surveys of capercaillie (*Tetrao urogallus*) in the Swiss Alps underestimated local abundance of the species as revealed by genetic analyses of non-invasive samples. Conserv Genet 11:33–44
Jones-Walters L, Çil A (2011) Biodiversity and stakeholder participation. J Nat Conserv 19:327–329
Kalniņš A (1943) Huntsmanship. Latvju grāmata, Rīga **(in Latvian)**
Kask E (1973) That the Capercaillie might continue to mount treetops. Eesti Loodus 16:208–210 **(in Estonian)**
Keskkonnaagentuur (2014) Yearbook Forest 2013. Estonian Environmental Agency, Tartu
Krüdener A (1898) Von der Auerhahnbalz in Livland, 1898. Das Waldwerk in Wort und Bild 24:306–309
Kuku J (1940) Let's extirpate hawks and raise game! Eesti Mets 20:286–288 **(in Estonian)**
Kumari E (1954) Birds of the Estonian S.S.R. Eesti Riiklik Kirjastus, Tallinn **(in Estonian)**
Kurki S, Nikula A, Helle P, Lindén H (2000) Landscape fragmentation and forest composition effects on grouse breeding success in boreal forests. Ecology 81:1985–1997
Kurlavičius P, Kuuba R, Lukins M, Mozgeris G, Tolvanen P, Angelstam P, Karjalainen H, Walsh M (2004) Identifying high conservation value forests in the Baltic States from forest databases. Ecol Bull 51:351–366
Lambeck RJ (1997) Focal species: a multi-species umbrella for nature conservation. Conserv Biol 11:849–856
Lazdinis I, Mavsar R, Weiss G, Lazdinis M (2009) Trends in marketing of non-market forest goods and services in the Baltic States. Balt For 15:65–75
Leivits M (2014) Characterisation of capercaillie metapopulation spatial structure by means of habitat suitability model. Publ Inst Geogr Univ Tartu 111:248–261
Logminas V (1960) Distribution of Capercaillie in Lithuania. Lietuvos Mokslų Akademijos darbai, Serija C 2(22):97–107
Logminas V (1962) Distribution, ecology and possibilities of the population management of some grouse species in Lithuania. PhD dissertation. Vilnius **(in Lithuanian)**
Lõhmus E (1984) Estonian forest site types. Eesti NSV Agrotööstuskoondise Info- ja juurutusvalitsus, Tallinn **(in Estonian)**
Lõhmus A (2011) Three-year periodicity in historical raptor-persecution data: an indication of vole cycles? Estonian J Ecol 60:155–164
Loudon H (1909) Isolierte Auerhühner. Neue Baltische Waidmannsblätter 13:302
MacMillan DC, Marshall K (2006) The Delphi process—an expert-based approach to ecological modelling in data-poor environments. Anim Conserv 9:11–19
Martenson A (1899) Wald, Wild und Jadg in den russischen Ostseeprovintzen. Neudamm
Miettinen J, Helle P, Nikula A, Niemelä P (2009) Changes in landscape-scale habitat selection of capercaillie (*Tetrao urogallus*) in managed north-boreal forest. Silva Fenn 43:595–608
Mihelsons H (1958) Bird protection in Latvian SSR. Saudzējiet un mīliet dabu. Latvijas PSR Zinātņu Akadēmijas izdevniecība, Rīga, pp 119–137 **(in Latvian)**
Mikoláš M, Svitok M, Tejkal M, Leitão PJ, Morrissey RC, Svoboda M, Seedre M, Fontaine JB (2015) Evaluating forest management intensity on an umbrella species: Capercaillie persistence in central Europe. For Ecol Manag 354:26–34
Mollet P, Badilatti B, Bollmann K, Graf RF, Hess R, Jenny H, Mulhauser B, Perrenound A, Rudmann F, Sachot S, Studer J (2003) Verbreitung und Bestand des Auerhuhns *Tetrao urogallus* in der Schweiz 2001 und ihre Veränderungen im 19. und 20. Jahrhundert. Ornithol Beob 100:67–86
Mollet P, Kéry M, Gardner B, Pasinelli G, Royle JA (2015) Estimating population size for capercaillie (*Tetrao urogallus* L.) with spatial capture-recapture models based on genotypes from one field sample. PLoS ONE 10:e0129020

Monroe AP, Edmunds DR, Aldridge CL (2016) Effects of lek count protocols on greater sage-grouse population trend estimates. J Wildl Manag 80:667–678
Nolcken B (1870) Hunting in Livonia. Zhurnal Moskovskogo Obschestvo okhoty 6:23–27 **(in Russian)**
Pakkala T, Pellikka J, Lindén H (2003) Capercaillie *Tetrao urogallus*: a good candidate for an umbrella species in taiga forests. Wildl Biol 9:309–316
Pregernig M (2014) Framings of science-policy interactions and their discursive and institutional effects: examples from conservation and environmental policy. Biodivers Conserv 23:3615–3639
Priednieks J, Strazds M, Strazds A, Petriņš A (1989) Latvian Breeding Bird Atlas 1980–1984. Zinātne, Rīga
Rametsteiner E, Mayer P (2004) Sustainable forest management and Pan-European forest policy. Ecol Bull 51:51–57
Renno O (ed) (1993) Estonian Bird Atlas. Valgus, Tallinn **(in Estonian, English summary)**
Roberge JM, Mikusiński G, Svensson S (2008) The white-backed woodpecker: umbrella species for forest conservation planning? Biodivers Conserv 17:2479–2494
Rolstad J, Wegge P (1989a) Capercaillie (*Tetrao urogallus*) populations and modern forestry: a case for landscape ecological studies. Finn Game Res 46:43–52
Rolstad J, Wegge P (1989b) Effects of logging on Capercaillie (*Tetrao urogallus*) leks. II. Cutting experiments in southeastern Norway. Scand J Forest Res 4:111–127
Rolstad J, Wegge P, Sivkov AV, Hjeljord O, Storaunet KO (2009) Size and spacing of grouse leks: comparing Capercaillie (*Tetrao urogallus*) and Black grouse (*Tetrao tetrix*) in two contrasting Eurasian boreal forest landscapes. Can J Zool 87:1032–1043
Romanov AN (1960) Some ecological features of the Capercaillie in relation to cuttings in northern taiga. Trudy Komi filiala AN SSSR 9:61–73 **(in Russian)**
Romanov AN (1988) The Capercaillie. Agropromizdat, Moskva **(in Russian)**
Rootsmäe I, Rootsmäe L (1983) Die Ornitologische Literatur Estlands. Tartu
San-Miguel-Ayanz J, Parviainen J, Schuck A, Bozzano M, Estreguil C, Koskela J, Lier M, Spielmann M, Van Brusselen J, Bücking W, Caudullo G, Larsson TB (2011) Criterion 4: maintenance, conservation and appropriate enhancement of biological diversity in forest ecosystems. State of Europe's Forests 2011: status and trends in sustainable forest management in Europe. Forest Europe, UN Economic Commission for Europe and FAO, Oslo, pp 65–97
Santangeli A, Hanski IK, Mäkelä H (2013) Integrating multi-source forest inventory and animal survey data to assess nationwide distribution and habitat correlates of the Siberian flying squirrel. Biol Conserv 157:31–38
Savchenko IA (2009) Forest management influence on the Hazel grouse and the Wood-grouse resources in the Central Siberia Subtaiga. Vestnik KrasGAU 9:117–121 **(in Russian)**
Sayer JA, Maginnis S (2005) Forests in landscapes: Expanding horizons for ecosystem forestry. In: Sayer JA, Maginnis S (eds) Forests in landscapes: ecosystem approaches to sustainability. The Earthscan Forestry Library, London, pp 177–192
Segura A, Castaño-Santamaría J, Laiolo P, Obeso JR (2014) Divergent responses of flagship, keystone and resource-limited bio-indicators to forest structure. Ecol Res 29:925–936
Sjöberg K (1996) Modern forestry and the Capercaillie. In: DeGraaf RM, Miller RI (eds) Conservation of faunal diversity in forested landscapes. Chapman & Hall, London, pp 111–135
Spies TA, McComb BC, Kennedy RSH, McGrath MT, Olsen K, Pabst RJ (2007) Potential effects of forest policies on terrestrial biodiversity in a multi-ownership province. Ecol Appl 17:48–65
Storch I (1997) The importance of scale in habitat conservation for an endangered species: the Capercaillie in Central Europe. In: Bissonette JA (ed) Wildlife and landscape ecology: effects of pattern and scale. Springer, New York, pp 310–330
Storch I (2007) Conservation status of grouse worldwide: an update. Wildl Biol 13(Suppl 1):5–12
Suchant R, Braunisch V (2004) Multidimensional habitat modelling in forest management: a case study using Capercaillie in the Black Forest, Germany. Ecol Bull 51:455–546
Summers RW (2007) Stand selection by birds in Scots pinewoods in Scotland: the need for more old-growth pinewood. Ibis 149(Suppl 2):175–182
Suter W, Graf RF, Hess R (2002) Capercaillie (*Tetrao urogallus*) and avian biodiversity: testing the umbrella-species concept. Conserv Biol 16:778–788
Teino J (1939) On game management in 1918–1938. Eesti Mets 19:33–37 **(in Estonian)**
Toll E (1987) Beiträge zu einer Avifauna Estlands unter besonderer Berücksichtigung der Insel Oesel. Holtland
Tyzenhauz K (1842–1846) General ornithology, with the description of birds of the world. Glucksberg, Wilno **(in Polish)**
Uliczka H, Angelstam P, Roberge JM (2004) Indicator species and biodiversity monitoring systems for non-industrial private forest owners: is there a communication problem? Ecol Bull 51:379–384

United States Fish and Wildlife Service (USFWS) (2011) Revised recovery plan for the northern spotted owl, *Strix occidentalis caurina*. US Fish Wildlife Service, Portland

Viht E (1985) The influence of human action on the Tetraonids population. Ornitol kogum 10:48–57 **(in Estonian, English summary)**

Viht E (1988) Grouse at Lahemaa National Park. Lahemaa uurimused 3:158–161

Viht E (1990) Status of tetraonid populations in Estonia. In: Proceedings of the 5th conference on the study and conservation of migratory birds of the Baltic Basin, vol 2. Zinātne, Rīga, pp 216–225

Viht E (1974) Changes in the numbers of the Tetraonidae in Estonia. Eesti Lood 17:349–350 **(in Estonian, English summary)**

Viht E, Randla T (2002) The Capercaillie: status of the Estonian population and its survival prospects. Hirundo Suppl 5:1–50

Wegge P, Rolstad J (2011) Clearcutting forestry and Eurasian boreal forest grouse: long-term monitoring of sympatric Capercaillie *Tetrao urogallus* and Black grouse *T. tetrix* reveals unexpected effects on their population performances. For Ecol Manag 261:1520–1529

Werner SR, Spurgeon JPG, Isaksen GH, Smith JP, Springer NK, Gettleson DA, N'Guessan L, Dupont JM (2014) Rapid prioritization of marine ecosystem services and ecosystem indicators. Mar Policy 50:178–189

Wood S, Scheipl F (2014) gamm4: generalized additive mixed models using mgcv and lme4. R package version 0.2-3. http://www.CRAN.R-project.org/package=gamm4

Young JC, Jordan A, Searle KR, Butler A, Chapman DS, Simmons P, Watt AD (2013) Does stakeholder involvement really benefit biodiversity conservation? Biol Conserv 158:359–370

Zawadzka D (2004) Directions in active protection of Capercaillie in Poland. Part I. History and present state. Sylwan 5:50–59 **(jn Polish, English summary)**

Zizas R (2015) The influence of forest structure on the Capercaillie (*Tetrao urogallus* L.) habitats and distribution in south-eastern Baltic Sea region. PhD Dissertation. Aleksandras Stulginskis University, Lithuanian Research Centre for Agriculture and Forestry